

CHG 569

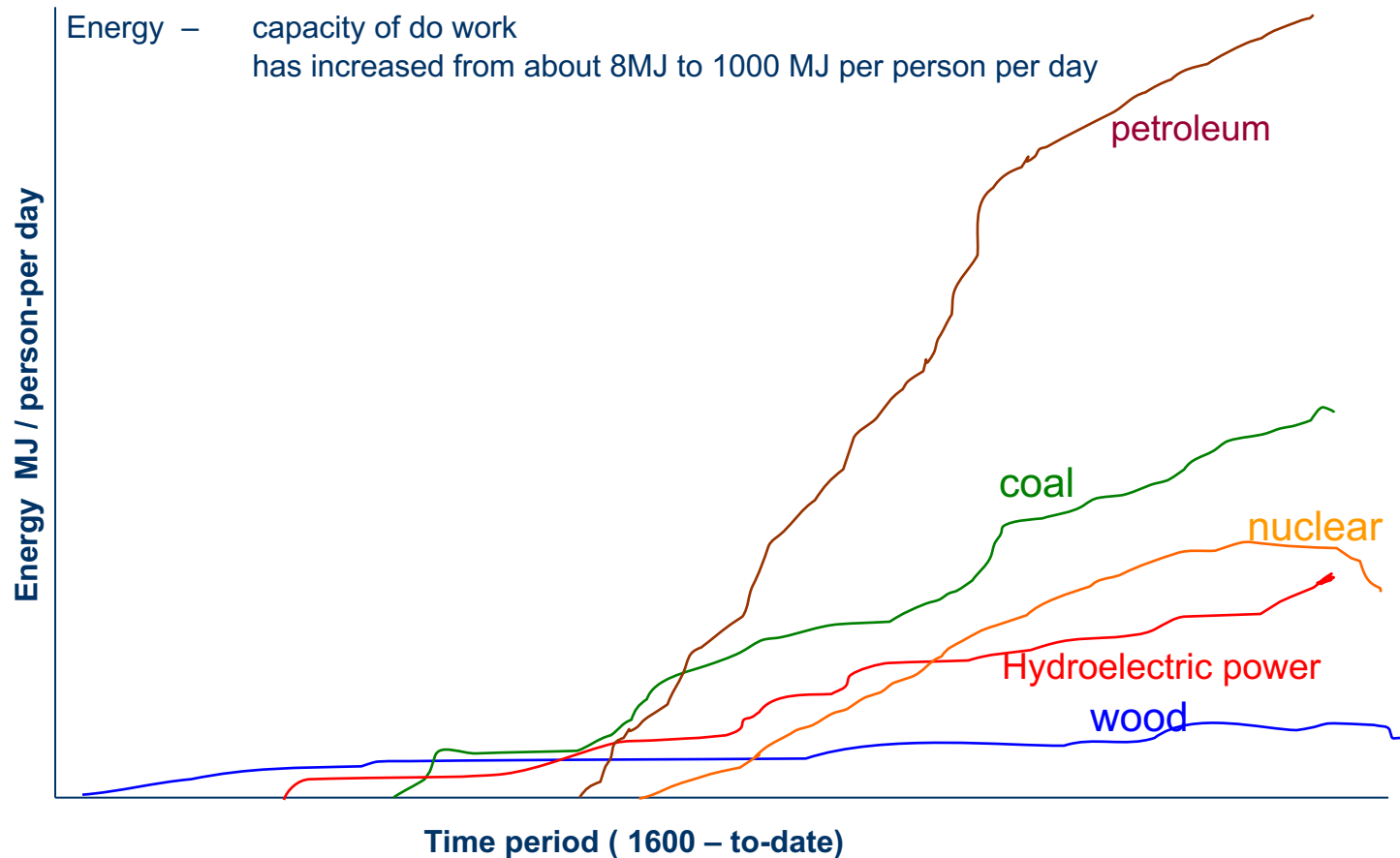
Environmental Pollution Engineering

Module III

(Energy & the Environment)



Energy and the Environment



Major sources of Energy- Nonrenewable

- Fossil fuel (coal, crude oil, gas)

Nonrenewable - rate of extraction >>>>>>> rate of formation

Chief source of environmental problems - combustion in industry or transportation gives rise to NO_x , CO_2 , SO_x , unburnt HCs

Coal, the dirtiest of the fossil fuels

gives 41% of the world's electricity (China - greatest user, followed by India. the USA)

Note that US has only



4.23% of the world's population
3% of the world's fossil but uses
25% of the world's fossil fuels
(herein lies the politics-



Many developed countries such as the USA & UK no longer want to rely on oil and gas imported from politically unstable parts of the world.)

Anyway,

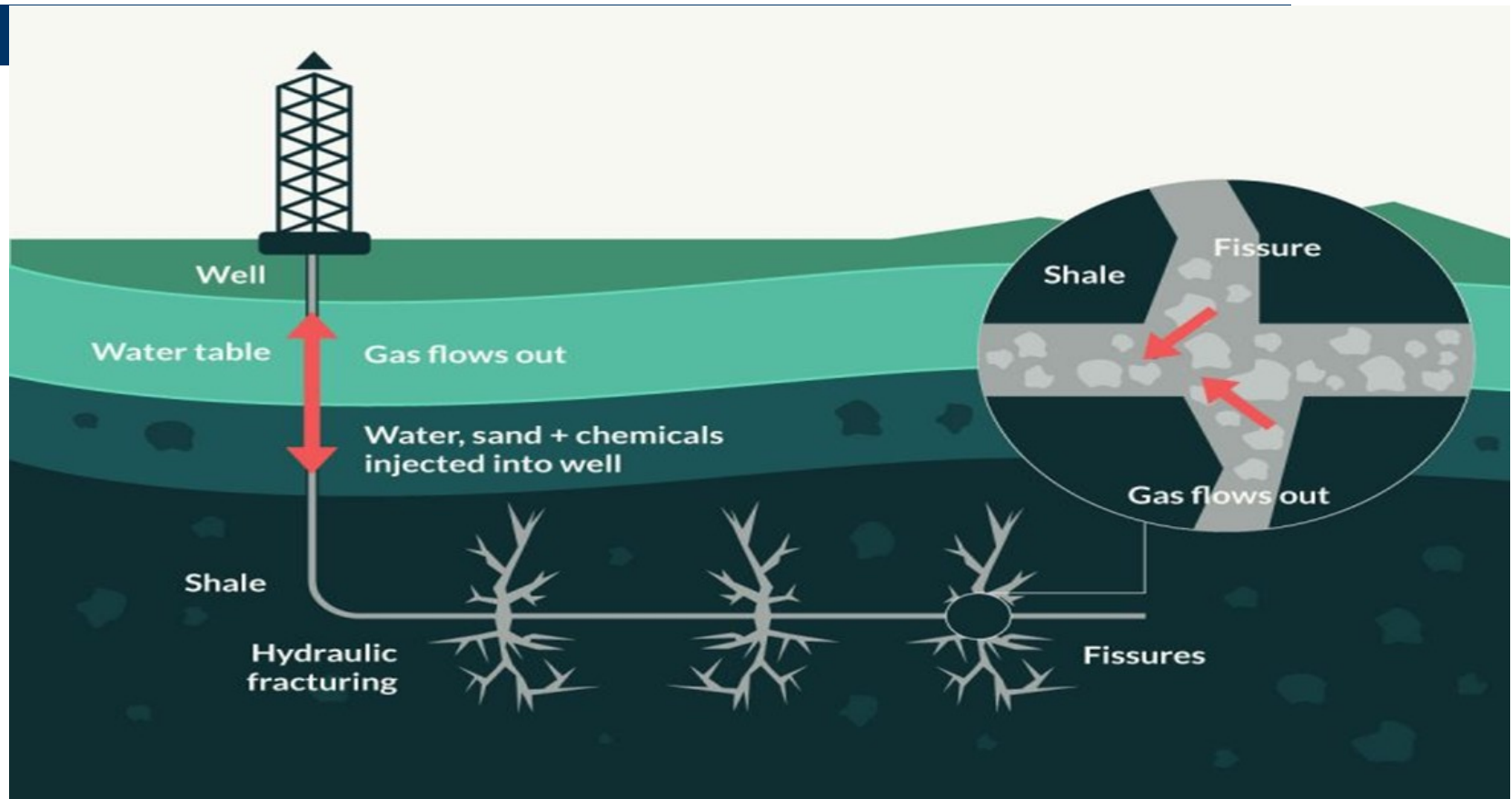
We have been made to believe that the current use of fossil fuel is not
Environmentally, Economically and Socially SUSTAINABLE

(THE 3PS OF Sustainable Development – People, Planet & Profit)
The Triple bottom line (TBL)

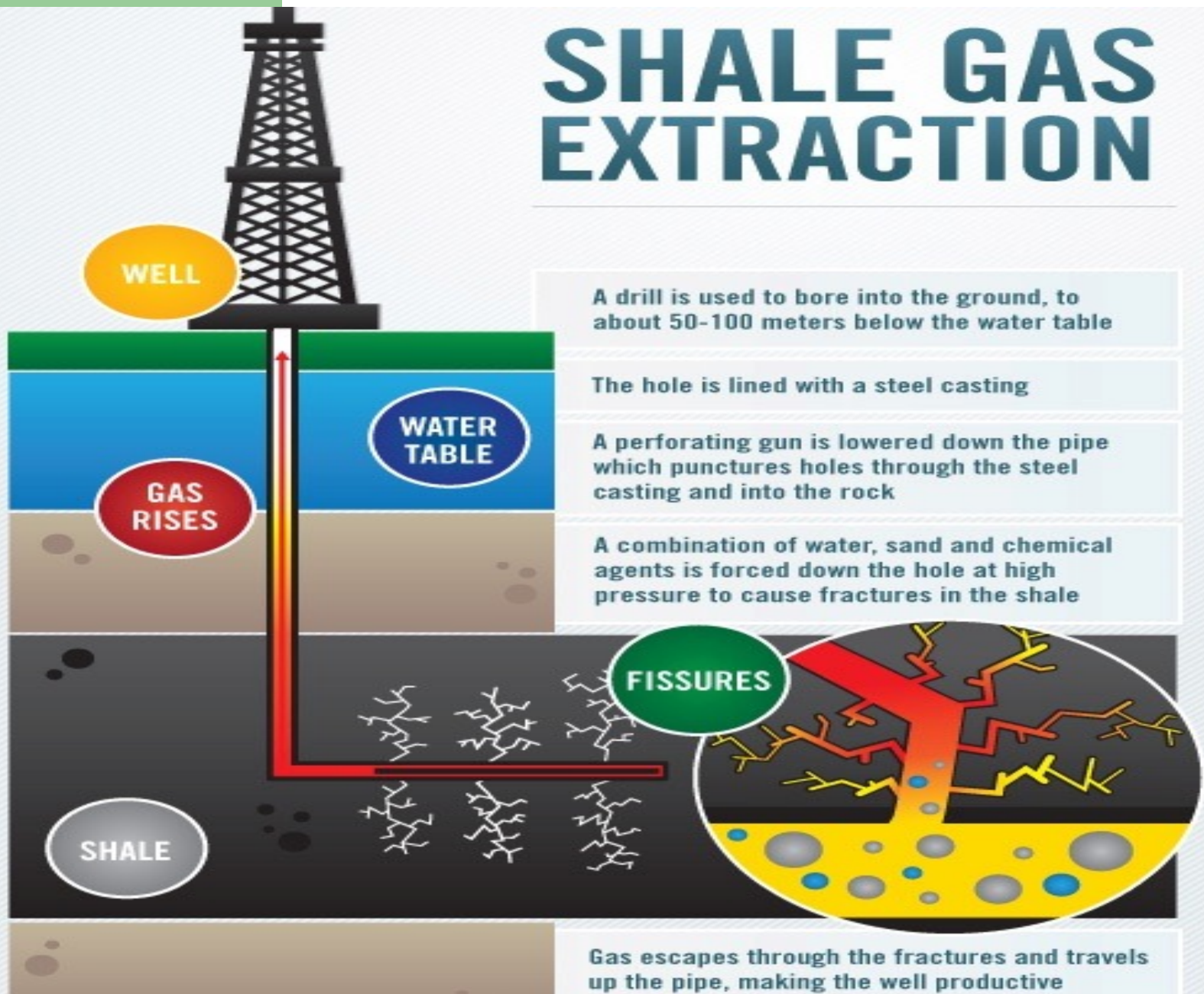
Energy independence/dominance for the US, 'thanks' to FRACKING

What is fracking?

Hydraulic fracturing (fracking) is the extracting oil and gas trapped in shale and other rock formations. It involves pumping large amounts of water down a well at high pressure, along with sand and chemicals. This fractures the rock and releases the gas or oil, which flows to the surface.



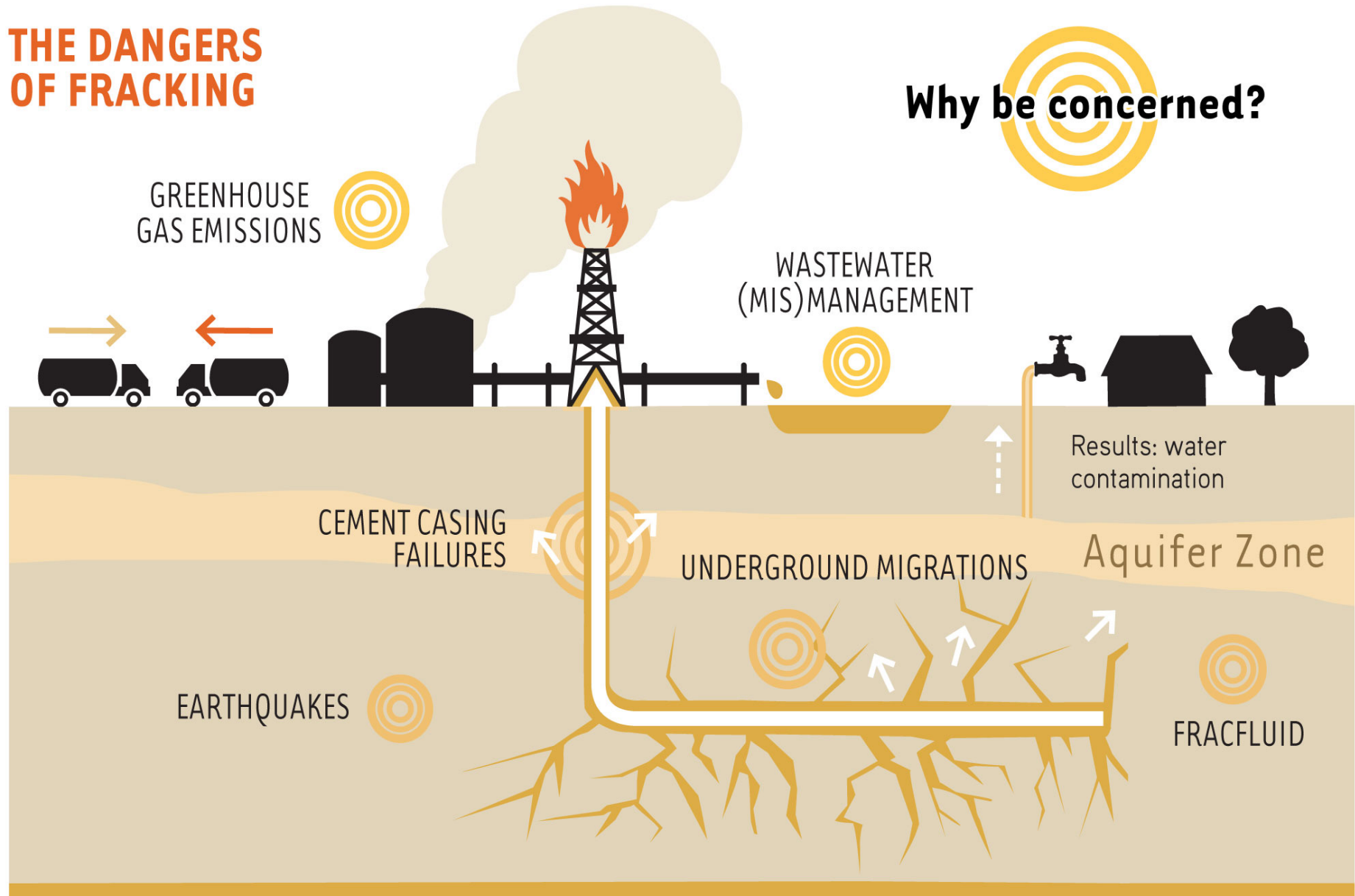
SHALE GAS EXTRACTION





THE DANGERS OF FRACKING

Why be concerned?



WHY IS FRACKING DANGEROUS?

1. Aquifer

To drill down to the shale, one must drill through the aquifer. These drills are known to leak and sometimes even explode, releasing chemicals into this precious source of water.

2. Chemicals

The 2011 U.S. House of Representatives investigative report states that out of 2,500 hydraulic fracturing products, more than 650 contain chemicals that are known carcinogens. One would think that the Safe Drinking Water Act - a Federal law - would make such willful contamination illegal, but it is not being applied to Fracking.

Also, most of these chemicals are not biodegradable. Once they are introduced to the aquifer, they will remain there forever.

3. Wastewater

Each gas well requires 1-8 million gallons of fresh water. The used water is one of the most hazardous wastes in the U.S., containing carcinogenic Fracking chemicals, remnant oil and hydrocarbons, biocides, as well as naturally occurring radioactive materials, like radon, which is heavier than air and sinks into the communities where people work and live.

4. Air Pollution

There are air emissions associated with Fracking, which include methane leaks originating from wells, as well as emissions from the diesel or natural gas-powered equipment such as compressors, drilling rigs, pumps etc..

HAS FRACKING BEEN BANNED ANYWHERE?

Yes, in Vermont, Governor Peter Shumlin and legislature made Vermont the first U.S. state to ban Fracking.

New Jersey, which lacks the pockets of natural gas that one finds in Pennsylvania, Texas, and New York has banned the treatment of Fracking wastewater.

It has also been banned in France, Bulgaria, and South Africa

Fracking may cause earthquakes due to the high pressure used to extract oil and gas from rock and the storage of excess wastewater on site.

Major sources of Energy - Nonrenewable

- Nuclear (Uranium- 235)

Nonrenewable - but there exist vast deposits in Australia etc.....

FEARS -

Enrichment produces toxic waste material

Radioactive wastes (spent rods currently stored in deep caves and mines)

Nuclear accidents - a very delicate (unstable) plant

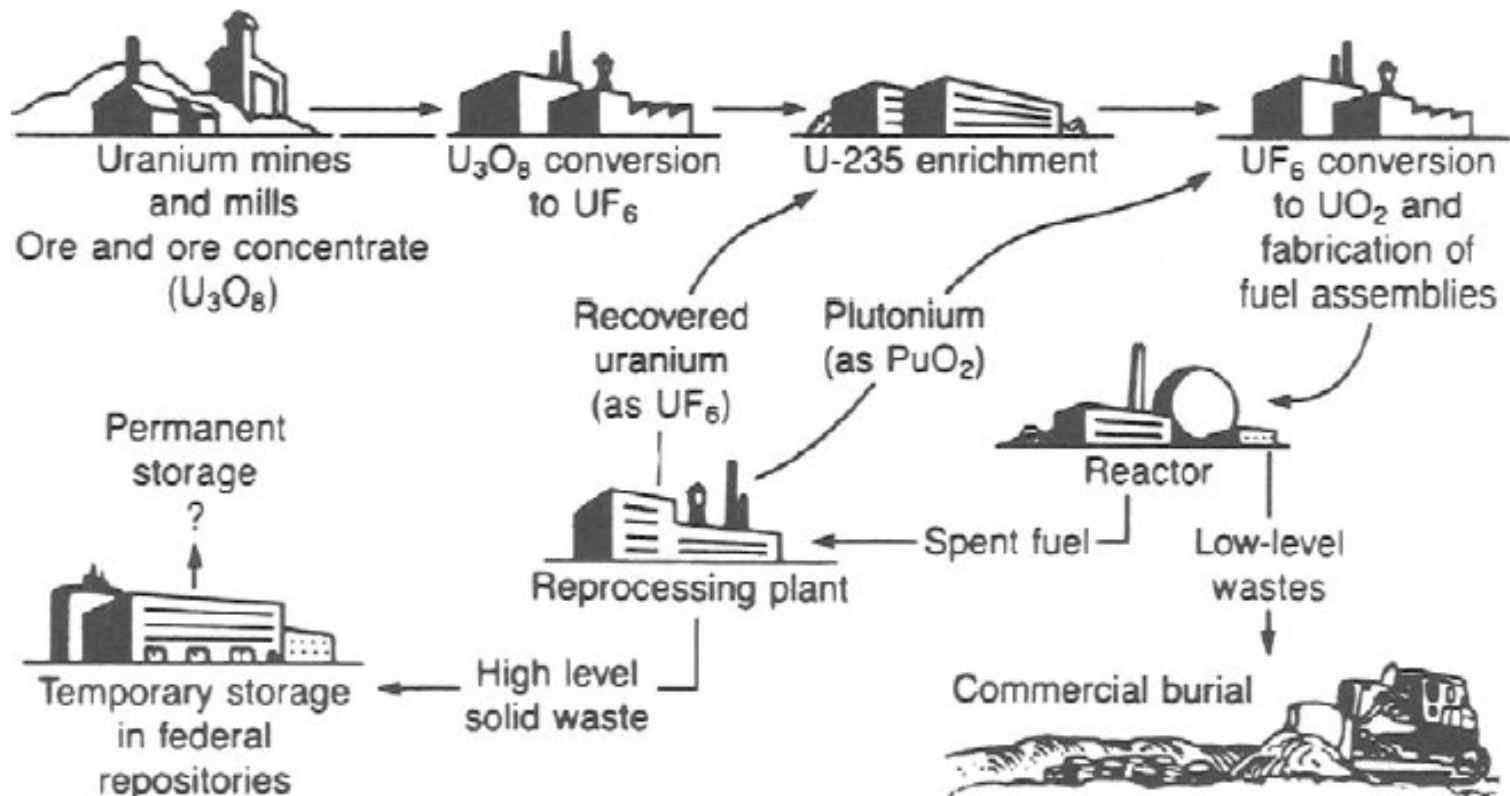
1979 Three Mile Island, Pennsylvania, USA

1986 Chernobyl, Ukraine (used to be part of the old USSR)

2011 Fukushima Daiichi, Japan

Nuclear weapon - especially by politically unstable countries

Nuclear fission produces much more energy than other sources, such as burning fossil fuels. 1kg of Uranium oxide produces heat equivalent to approximately 16 metric tonnes (16,000kg) of coal



The Nuclear Fuel cycle

Options for a Sustainable future

- **Energy Efficiency** — meeting up with increasing demand in the face of dwindling supplies
 - Getting more from less**
 - **Energy conservation** - use less energy e.g. LED & fluorescent replace incandescent bulbs
Switch off lights/Ac when leaving a room
Walk/ bike/ bus/ carpooling vs. hrs in traffic
Natural lighting and ventilation
solar paneling /inverters instead of generators
Use less water etc etc etc
 - **Increased efficiency** — equipment design to yield more energy from a given input energy
 - **Cogeneration** - Capture and use waste heat vs. release to the atmosphere - **CHP**

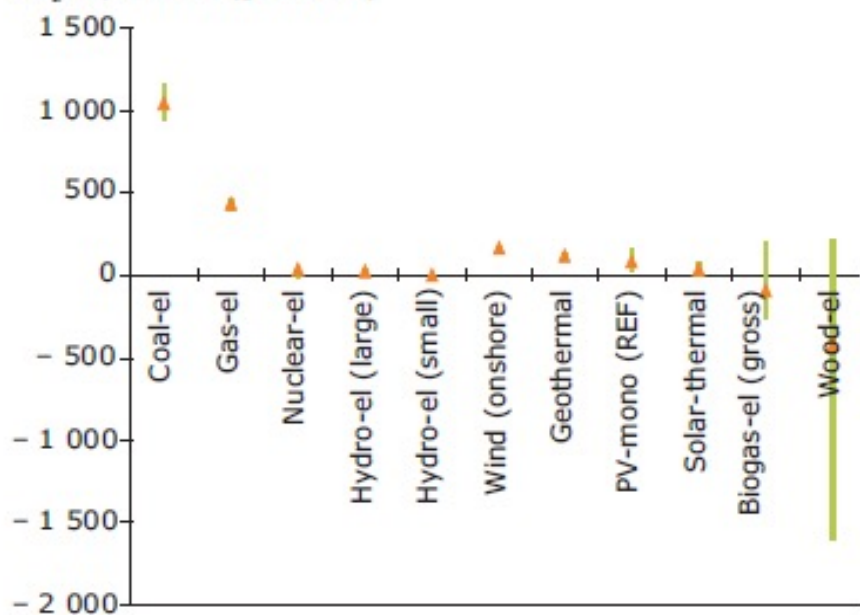
Options for a Sustainable future

- Alternative energy sources- Renewable

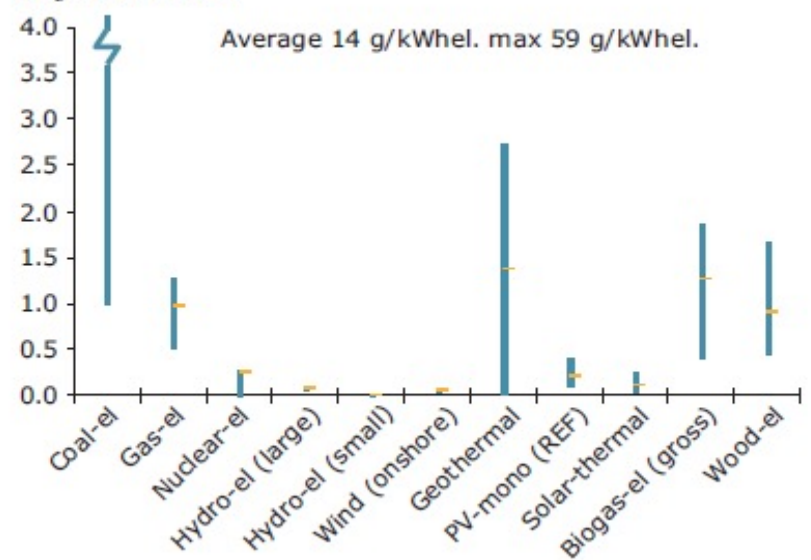
- | | |
|---------------------------|---|
| ➤ Biofuel | Fuel from biomass |
| ➤ Biomass | Organic matter |
| ➤ Geothermal | Hot formations used to produce heat |
| ➤ Hydroelectricity | Harnessing for use the energy in water |
| ➤ Solar energy | Electromagnetic radiation from the sun |
| ➤ Tidal power | Energy associated with the rise and fall of tidal waves |
| ➤ Wave power | Energy associated with ocean surface waves |
| ➤ Wind power | Energy from moving air |

LCA emissions from different energy sources

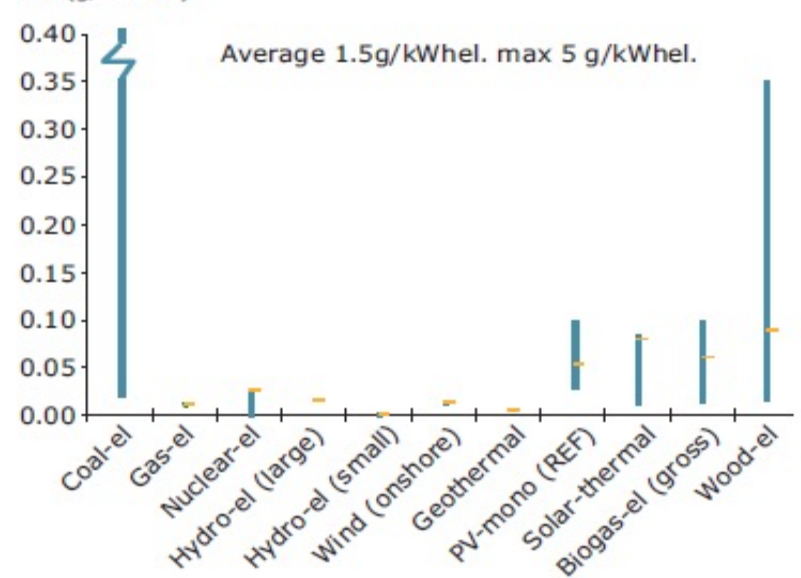
CO₂-equivalent (g/kWhel.)

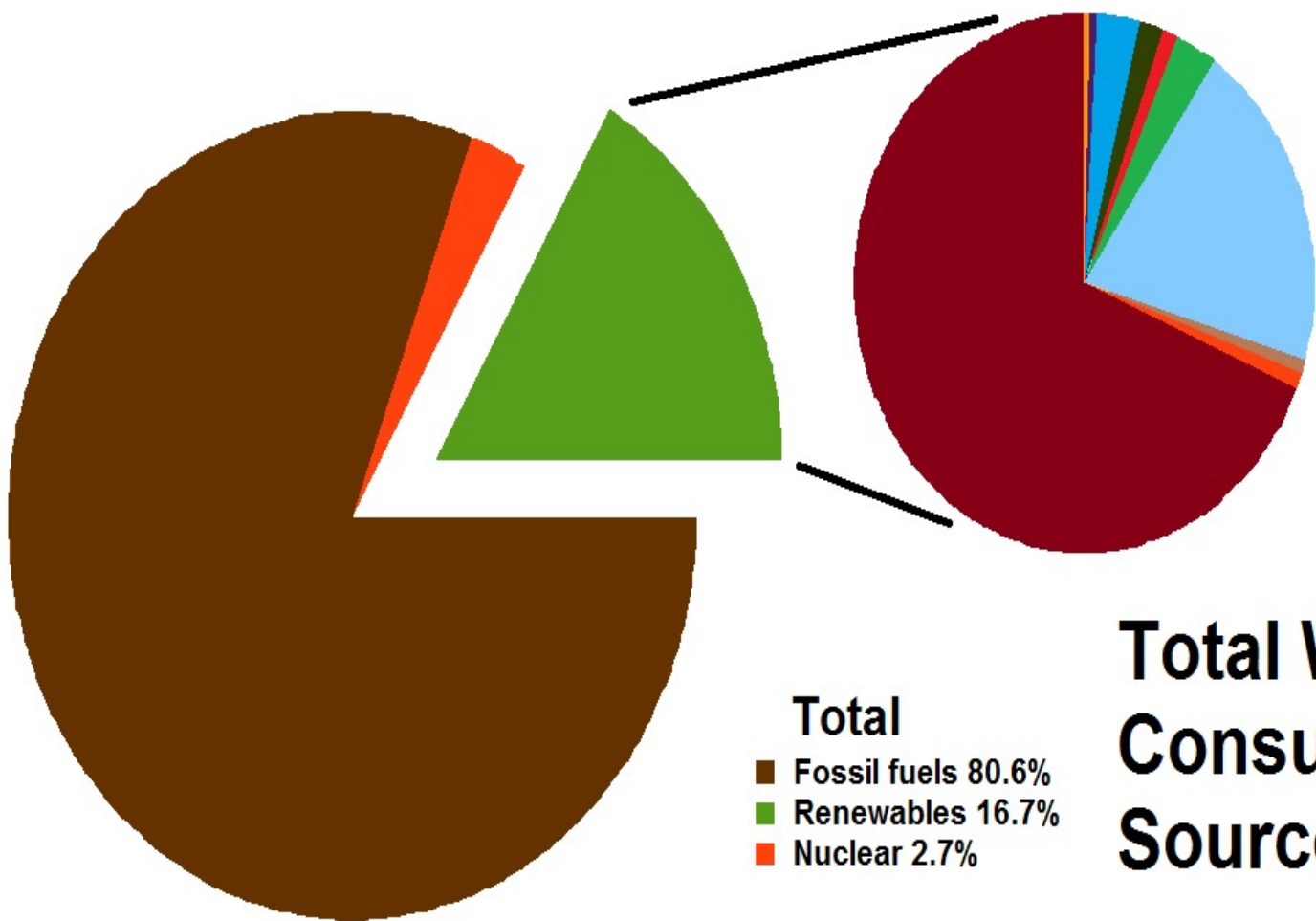


SO₂ eq. (g/kWhel.)



PM (g/kWhel.)

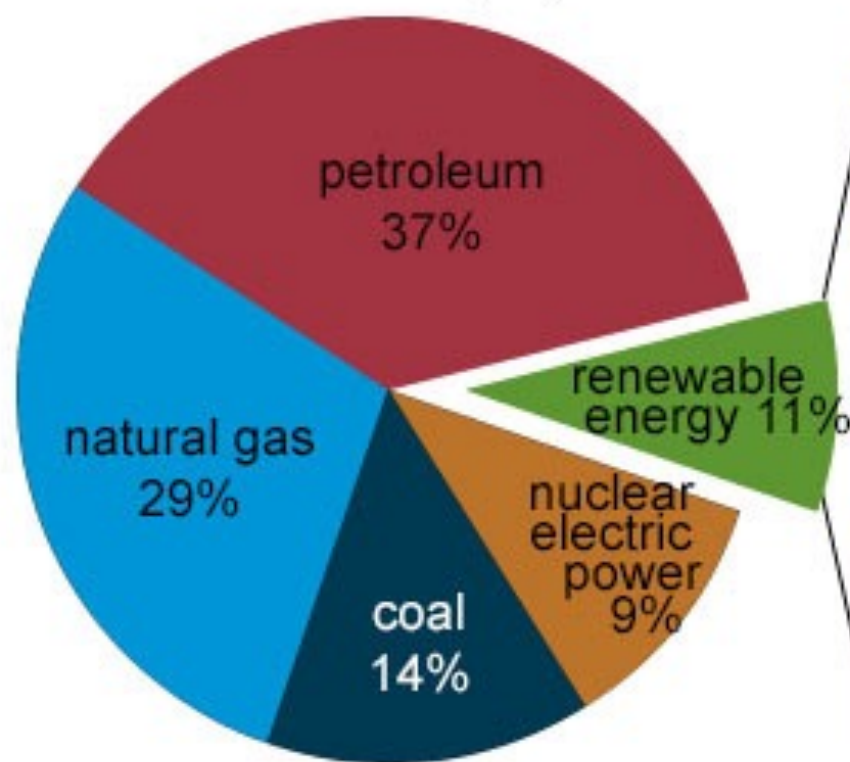




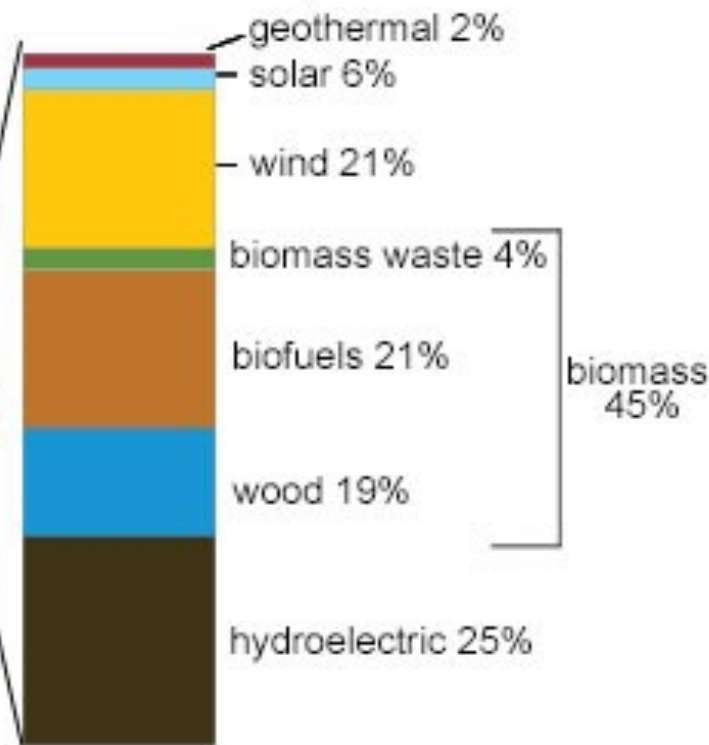
Total World Energy Consumption by Source (2010)

U.S. energy consumption by energy source, 2017

Total = 97.7 quadrillion
British thermal units (Btu)



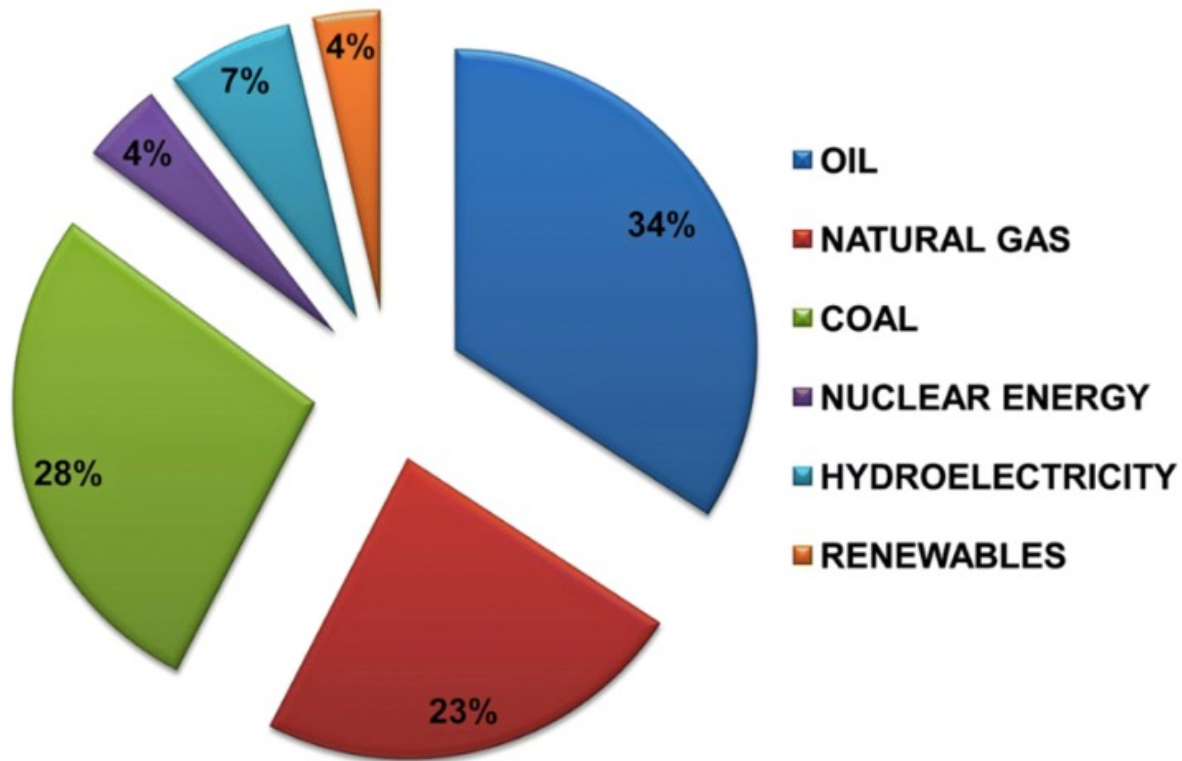
Total = 11.0 quadrillion Btu



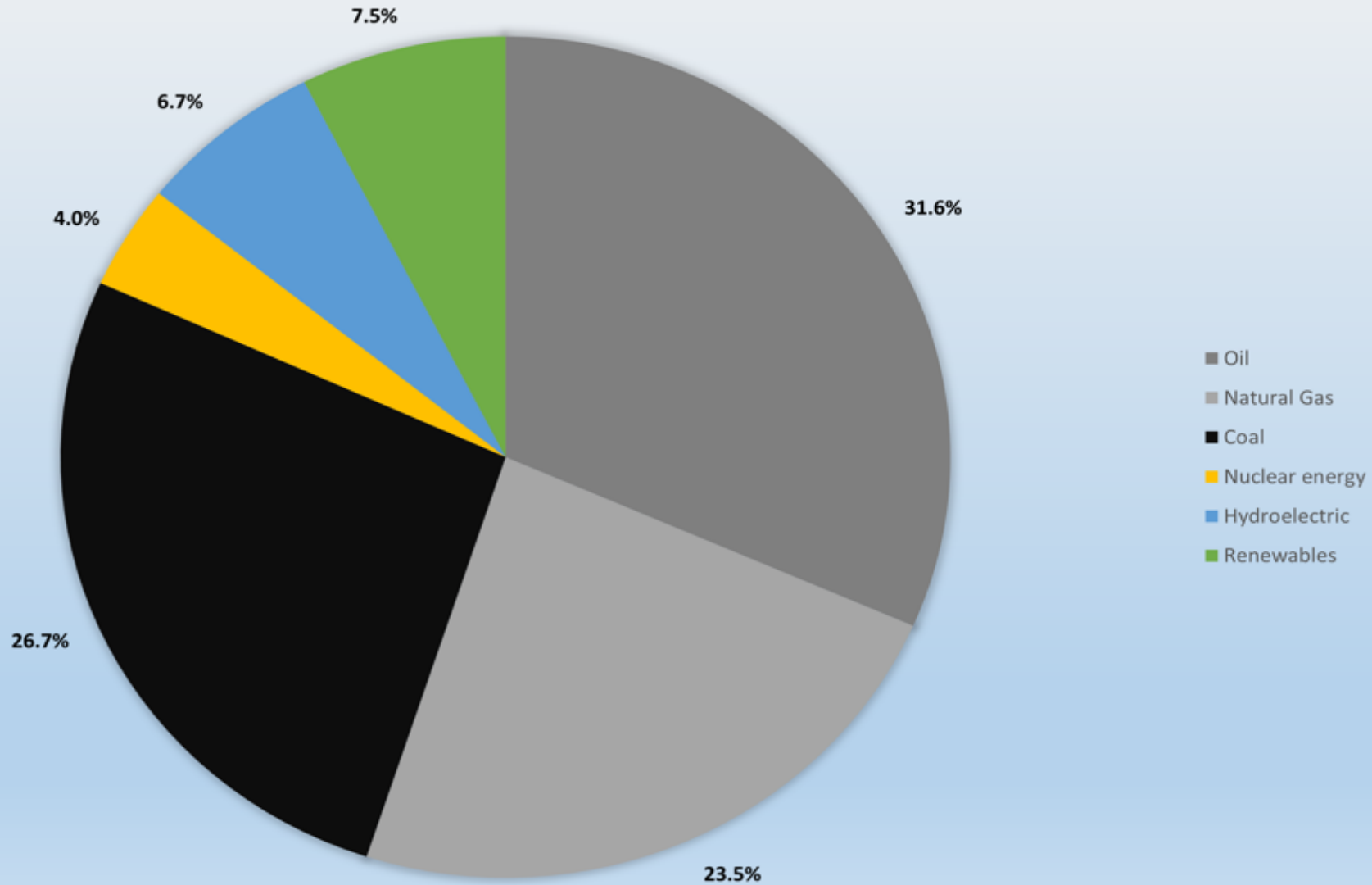
Note: Sum of components may not equal 100% because of independent rounding.
Source: U.S. Energy Information Administration, *Monthly Energy Review*, Table 1.3 and 10.1, April 2018, preliminary data



Global energy consumption 2018



PRIMARY GLOBAL ENERGY CONSUMPTION 2022



Data source: 2023 Statistical Review of World Energy

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- **Fossil fuels still provide around 70% of the world's electrical power and 95% of the world's total energy demand for heating, transport, electricity generation, etc. etc., and might remain so for a very long time**

Energy & the Environment

The relationship between energy and the environment is closely interconnected. Energy production and consumption have significant impacts on the environment. For instance, burning fossil fuels (coal, oil, and natural gas) releases greenhouse gases, such as carbon dioxide, contributing to global warming and climate change. It also leads to air and water pollution, affecting both human health and ecosystems. On the other hand, renewable energy sources like wind, solar, and hydroelectric power offer cleaner alternatives that have much lower environmental impacts. However, even renewable energy can have some negative effects on the environment, such as habitat disruption from hydroelectric dams or land use for solar farms. Thus, the challenge lies in balancing energy needs with environmental protection, striving for sustainable energy solutions that minimize harm to the environment.